INCH-POUND
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MILITARY SPECIFICATION SHEET

RELIEF VALVE OPERATING CHARACTERISTICS VS. MAXIMUM OPERATING PRESSURE FOR LIQUID SERVICE

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The maximum operating pressure curve shall be used to identify the relationship the designer needs to consider regarding the differential between the nominal set pressure of the valve and the operating pressure of the vessel/system (see figure 1). The curve was established to distinguish the margin between operating pressure and the relief valve settings such as accumulation, reseat, and nominal set pressure. It is used to minimize operating difficulty within the system between the control valve and the relief valve by providing an adequate differential for each liquid service application to stay within.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 4820

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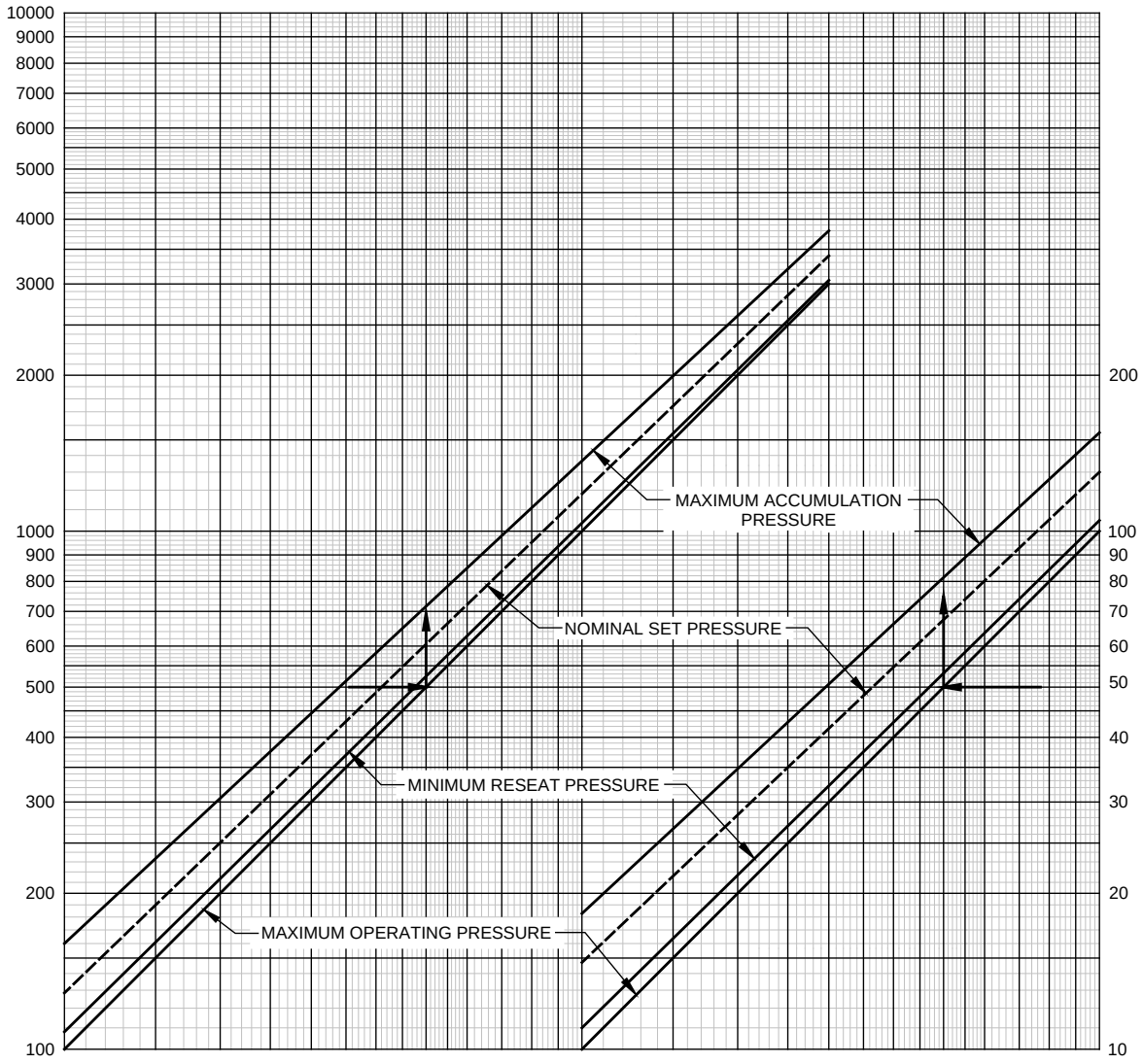

FIGURE 1. Maximum operating pressure – pounds per square inch gage (PSIG).

NOTE: Figure 1 is a log-log graph. The units of the vertical ordinates are PSIG. The right-hand vertical ordinate covers 10 to 100 PSIG, and the left-hand vertical ordinate covers 100 to 3000 PSIG. Figure 1 is used in the following way:

1. Find the desired maximum operating pressure on the left or right vertical ordinate.

2. Find the point on the graph where the desired maximum operating pressure intersects the line called "MAXIMUM OPERATING PRESSURE".

3. Moving vertically up from the point of intersection, identify the minimum reseat pressure, the nominal set pressure, and the maximum accumulation pressure.

CHANGES FROM PREVIOUS ISSUE. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Preparing activity:
Navy – SH
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